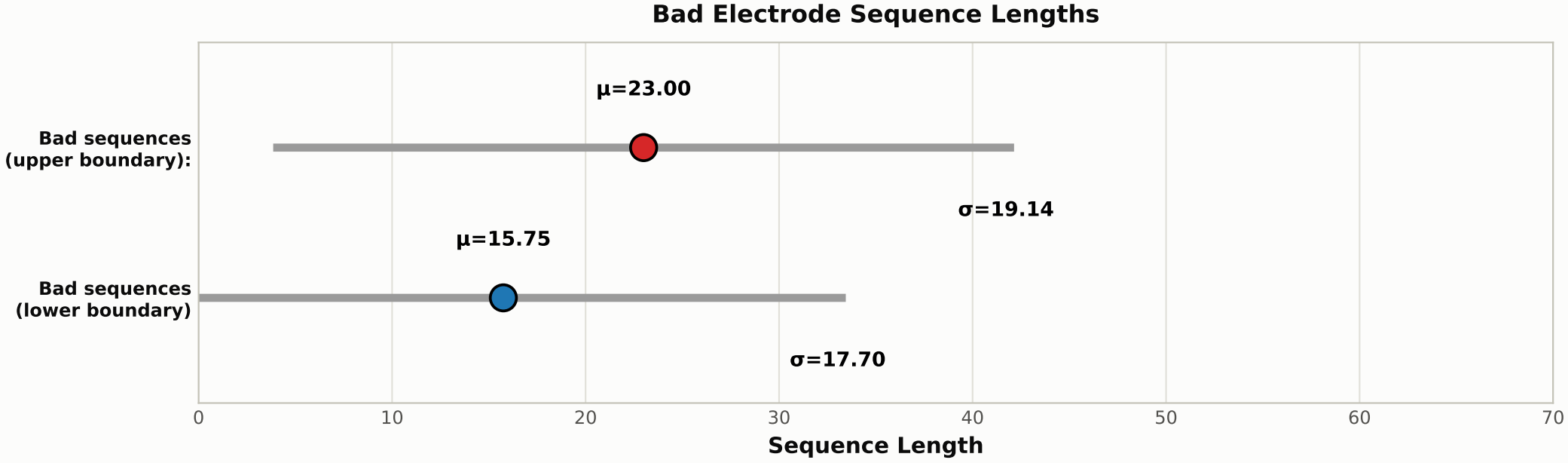
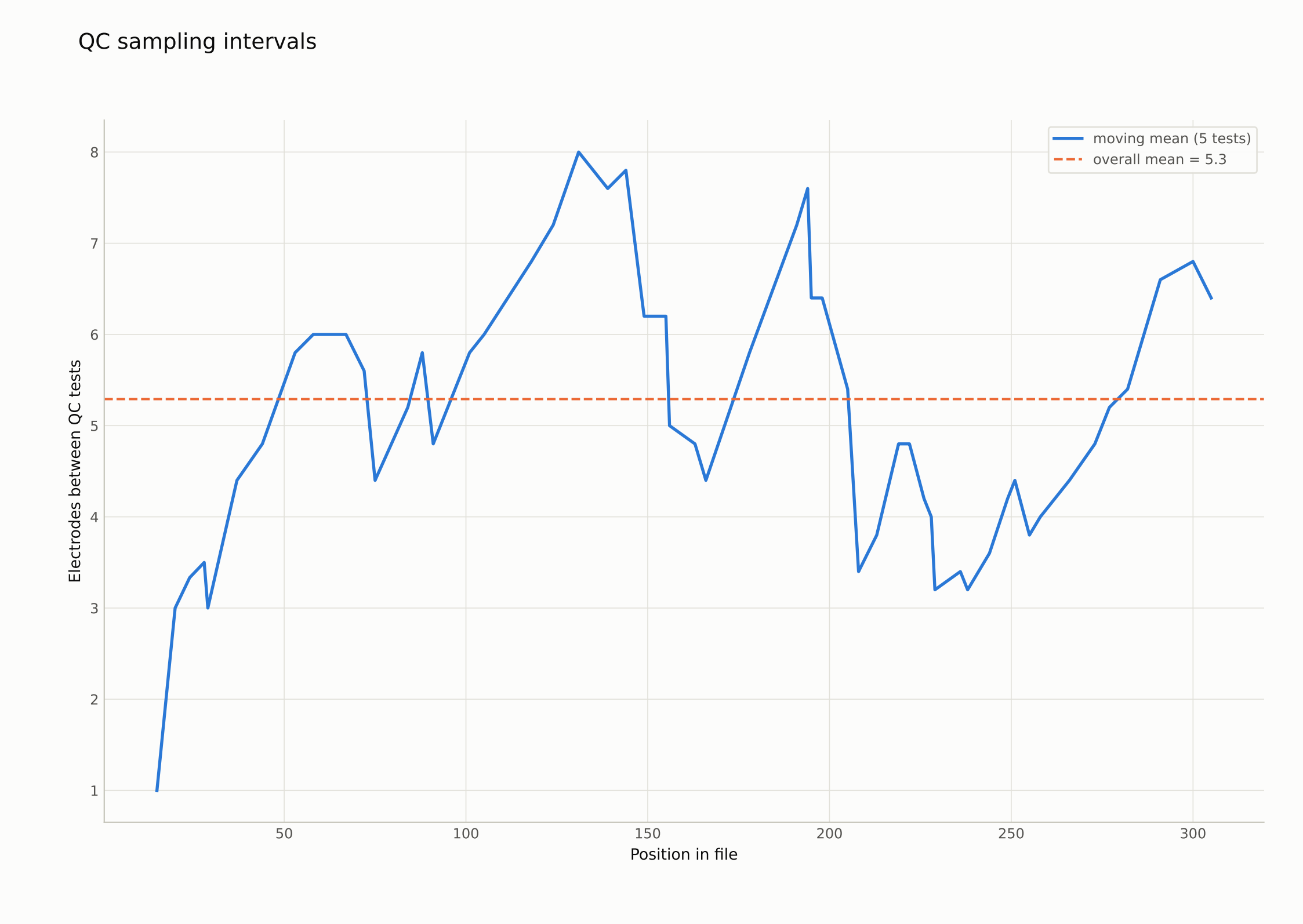
Sampling frequency — summary

```
Number of Gen4.1 electrodes produced: 373 (B13862 – B14349)

STATUS BREAKDOWN   (291 decided, 82 still untested)
  Released                  158    54.3%
  Scrapped                   77    26.5%
  Tested and Released        41    14.1%
  Tested and Scrapped        15     5.2%

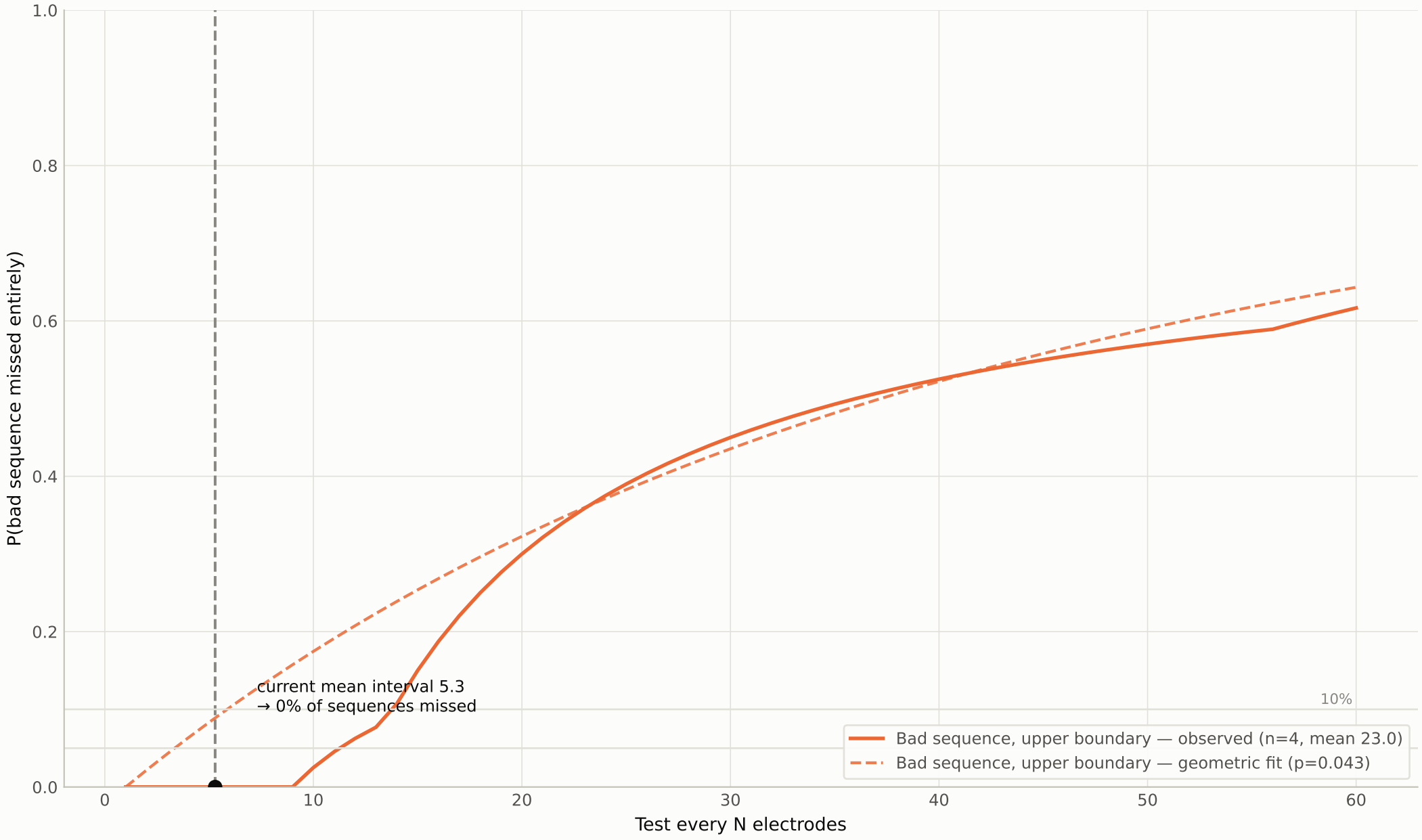
QC TESTING
  QC tests performed         56    15.0% of electrodes
    passed                   41
    failed                   15
  Mean interval             5.3 electrodes between tests
  Median interval           5.0
  Longest gap                13
```




QC sampling intervals
moving mean (5 tests)
overall mean = 5.3
Electrodes between QC tests
Position in file
1
2
3
4
5
6
7
8
50
100
150
200
250
300

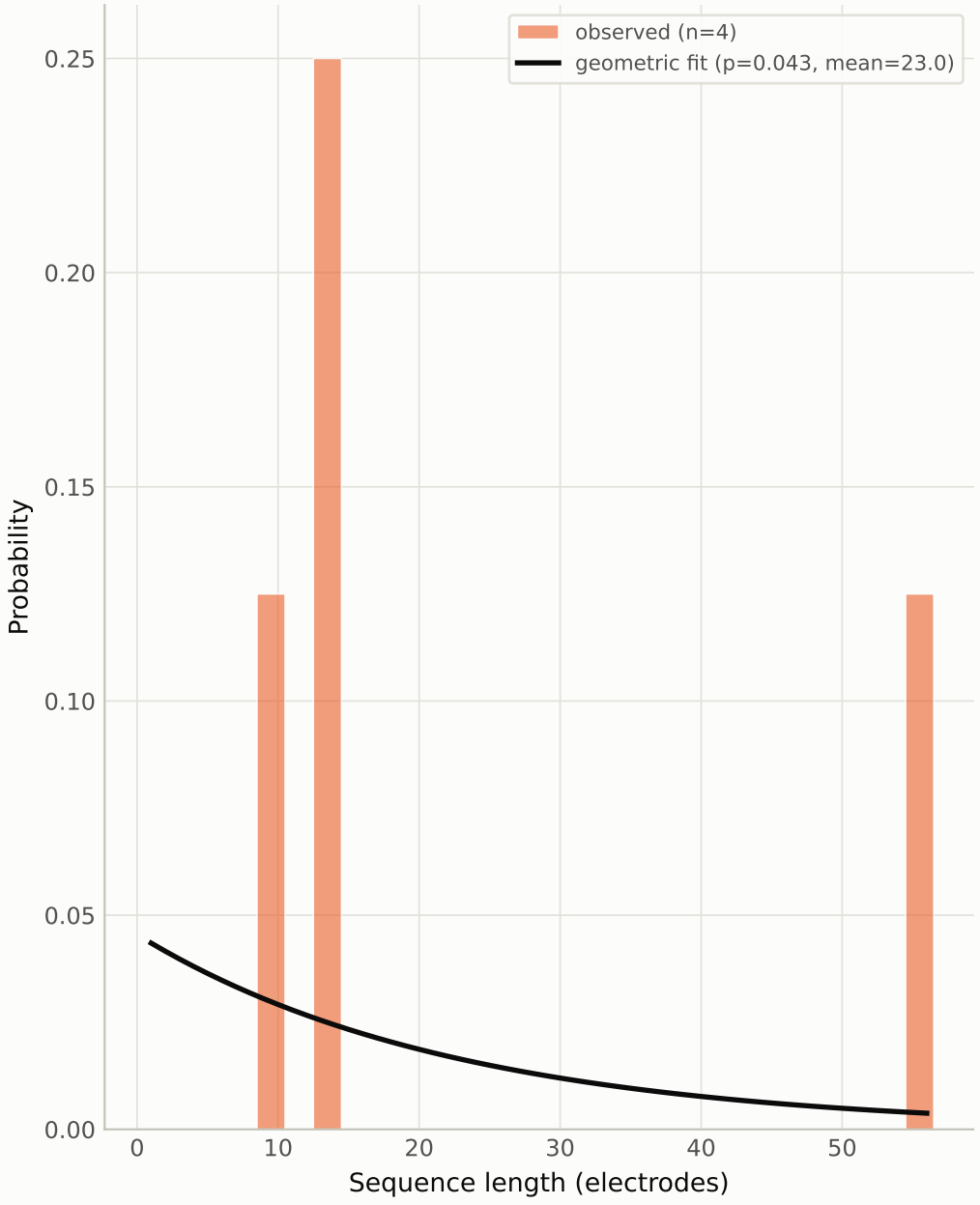
How often must we test?

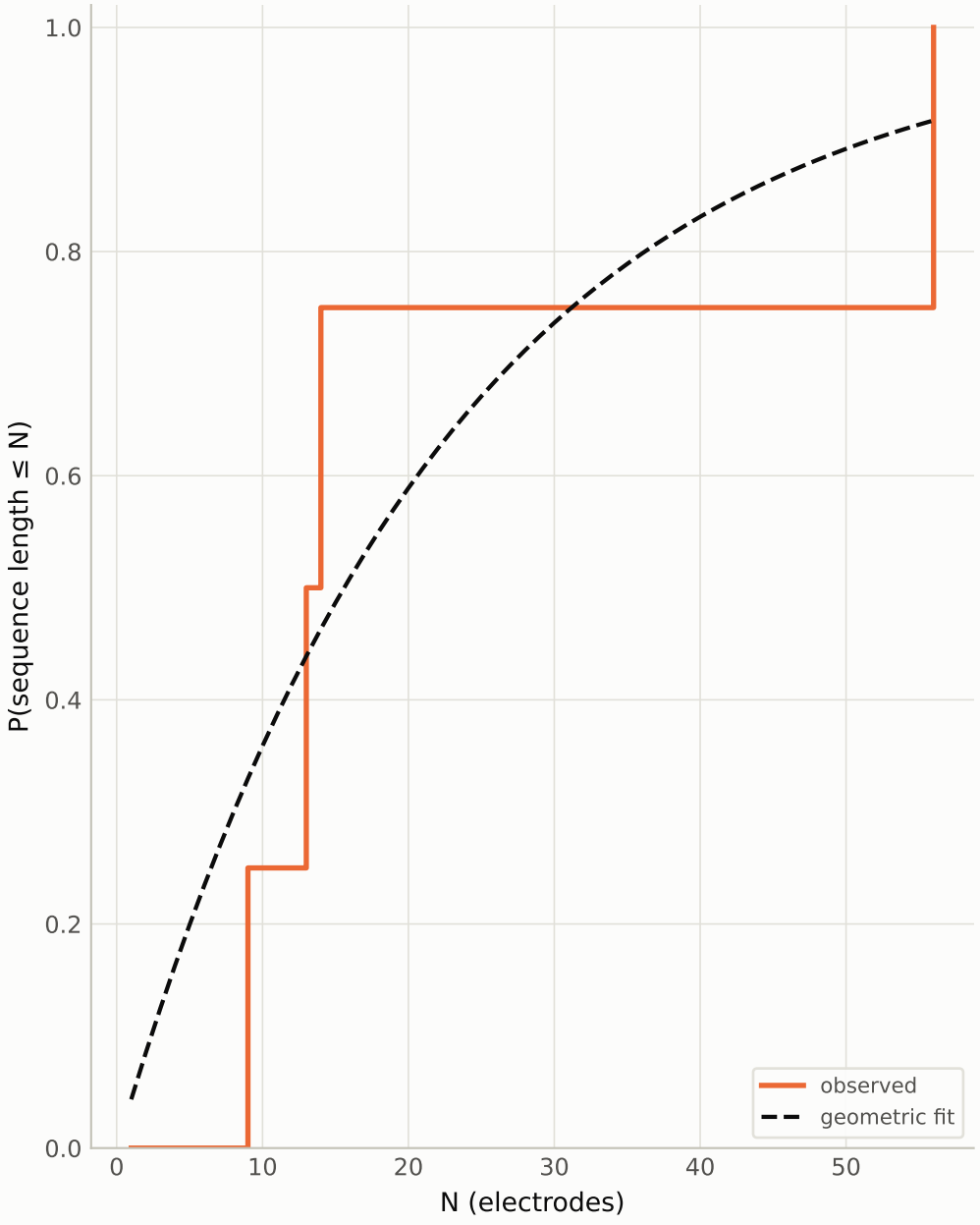
Probability a bad sequence starts and ends between two consecutive tests, so nothing catches it.



Sequence-length distribution — bad sequences, upper boundary

Window between the passing test before and the passing test after each failure episode.





QC failure episodes (1–4 of 4)

first fail	last fail	fails	confirmed	suspected	passed before	passed after
B14095	B14154	9	46	56	B14088	B14159
B14165	B14172	3	8	14	B14159	B14174
B14175	B14182	2	8	9	B14174	B14184
B14234	B14234	1	1	13	B14229	B14243

How to read this, and what to watch out for

Method

Bad electrodes arrive in sequences, not at random, so the question is a comparison of two lengths: how long a bad sequence lasts, versus how far apart the QC tests are. A sequence is only missed if it starts and finishes between two consecutive tests.

Two lengths per episode

Lower boundary = first to last failed test. Upper boundary = the whole window between the passing test before and the passing test after, which is the part nobody actually checked. The upper boundary is the honest number; the lower one understates the sequence.

Two populations

QC episodes use only genuine QC results ('Tested and …'). Final-status sequences use every electrode ultimately marked Scrapped, including those condemned by inference rather than by test. Check whether they agree before treating them as separate evidence.

Geometric, not normal

Sequence lengths are strictly positive and right-skewed, so a normal fit would place mass below zero. The geometric distribution is the natural model here and gives P(length ≤ N) directly.

Caveats

- 115 marking numbers between B13862 and B14349 are absent from the file. Lengths are counted in file positions, so if those electrodes exist but were not exported, every sequence length here is an undercount in real production terms.

- The CSV is not sorted by date — marking order and timestamp order disagree. This analysis uses marking order as the production sequence.

- The final-status bad sequences are IDENTICAL to the QC upper boundaries, electrode for electrode. The scrapped ranges were evidently assigned by the same 'condemn everything between two passing tests' rule, so they are not independent evidence — they restate